Power BI & Business Intelligence

Week 5 Assignment: Superstore Sales Dashboard

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InternNova Internship | Week 5
Submission date: 18 October 2026

Task 1: Introduction & Data Import

1.1 What is Power BI?

Power BI is a Microsoft business intelligence tool. It connects to data, cleans it, builds a model, calculates measures with DAX and shows the results in interactive reports and dashboards. Businesses use it to understand sales, profit and customers and to take better decisions.

1.2 Dataset used

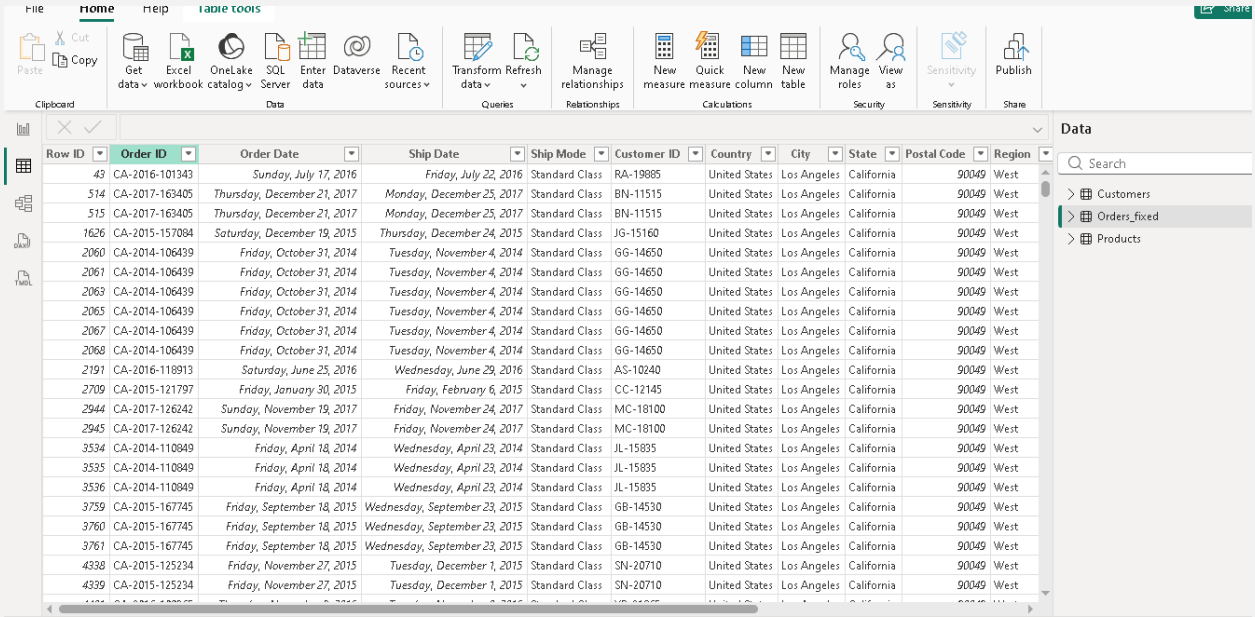
Dataset: Sample Superstore (9,994 rows). It holds retail orders with sales, quantity, discount, profit, customer, product and region details. For a proper star schema the data was split into three CSV files:

Table	Type	Main columns
Orders	Fact	Order ID, Order Date, Ship Date, Customer ID, Product ID, Region, Sales, Quantity, Discount, Profit
Customers	Dimension	Customer ID, Customer Name, Segment
Products	Dimension	Product ID, Category, Sub-Category, Product Name

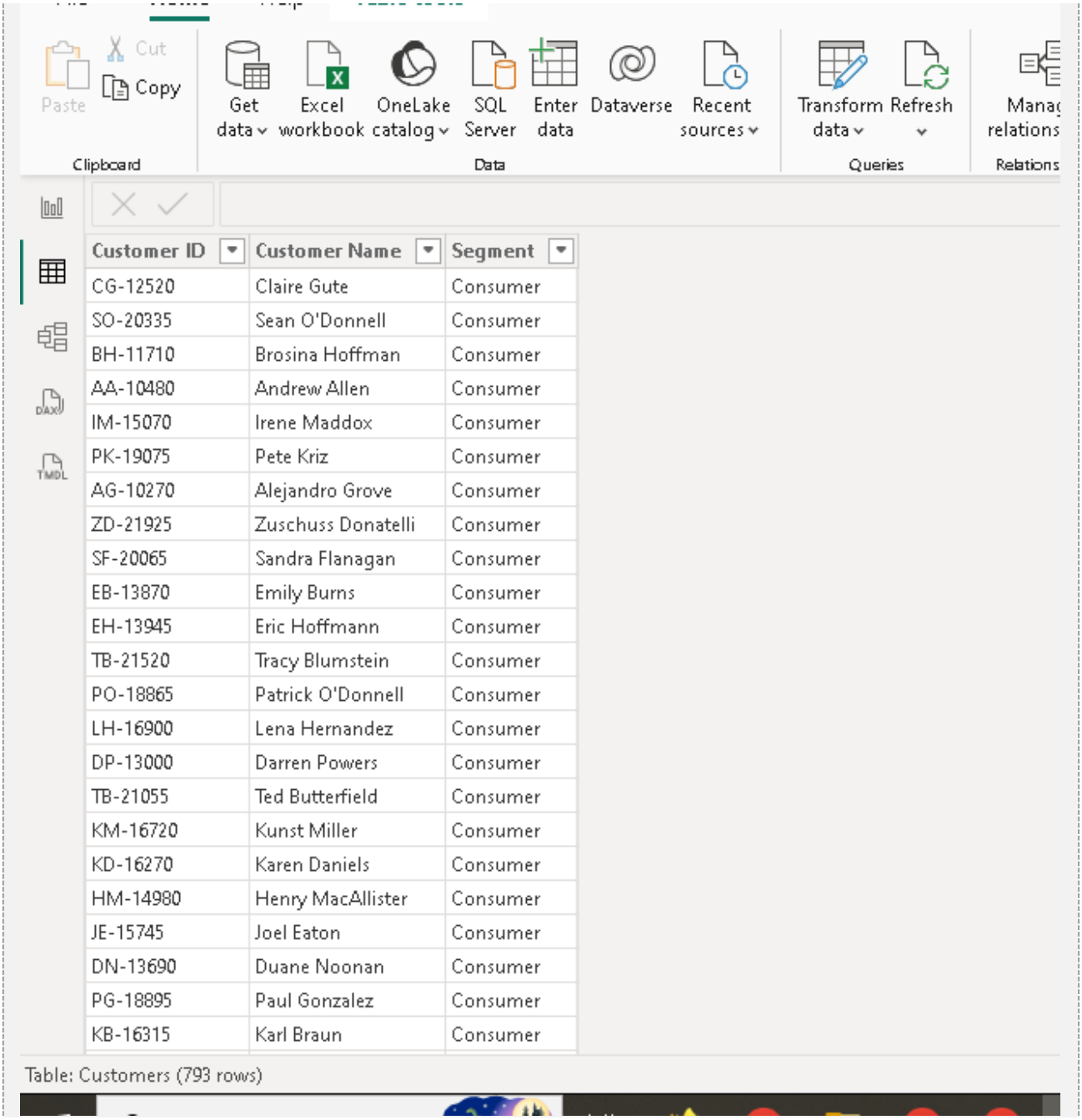
1.3 Import steps

1. Open Power BI Desktop, then Home > Get data > Text/CSV.
2. Select Orders.csv and click Transform Data (Load was not used directly).
3. Repeat for Customers.csv and Products.csv.
4. Check that the three tables are listed in the Queries pane.

[Screenshot here: Get data > Text/CSV window]

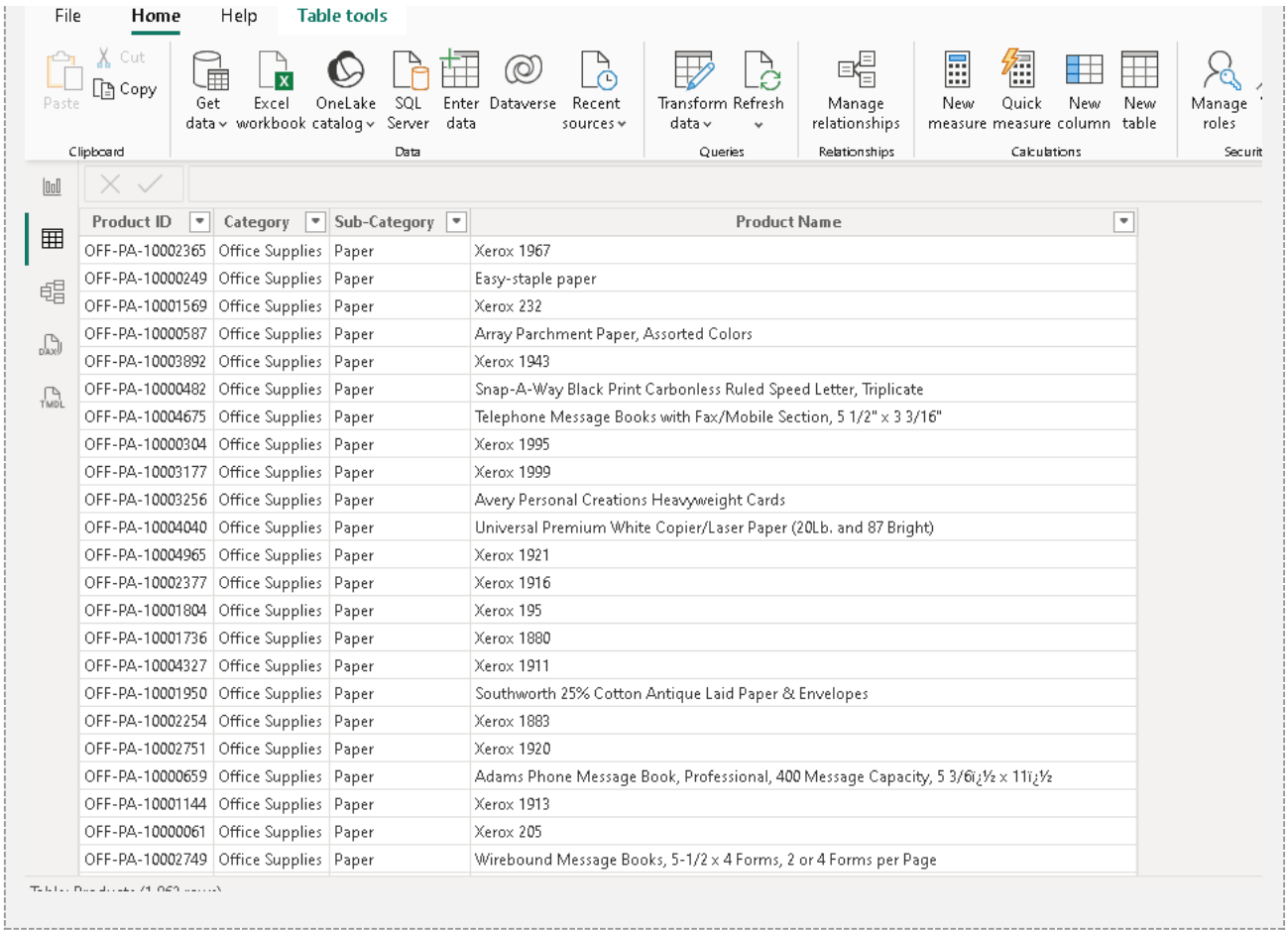


[Screenshot here: Imported tables in Power Query / Data pane }



Customer ID	Customer Name	Segment
CG-12520	Claire Gute	Consumer
SO-20335	Sean O'Donnell	Consumer
BH-11710	Brosina Hoffman	Consumer
AA-10480	Andrew Allen	Consumer
IM-15070	Irene Maddox	Consumer
PK-19075	Pete Kriz	Consumer
AG-10270	Alejandro Grove	Consumer
ZD-21925	Zuschuss Donatelli	Consumer
SF-20065	Sandra Flanagan	Consumer
EB-13870	Emily Burns	Consumer
EH-13945	Eric Hoffmann	Consumer
TB-21520	Tracy Blumstein	Consumer
PO-18865	Patrick O'Donnell	Consumer
LH-16900	Lena Hernandez	Consumer
DP-13000	Darren Powers	Consumer
TB-21055	Ted Butterfield	Consumer
KM-16720	Kunst Miller	Consumer
KD-16270	Karen Daniels	Consumer
HM-14980	Henry MacAllister	Consumer
JE-15745	Joel Eaton	Consumer
DN-13690	Duane Noonan	Consumer
PG-18895	Paul Gonzalez	Consumer
KB-16315	Karl Braun	Consumer

Table: Customers (793 rows)


File
Home
Help
Table tools
Cut
Copy
Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
Quick measure
New column
New table
Manage roles
Clipboard
Data
Queries
Relationships
Calculations
Product ID
Category
Sub-Category
Product Name
OFF-PA-10002365 Office Supplies Paper Xerox 1967
OFF-PA-10000249 Office Supplies Paper Easy-staple paper
OFF-PA-10001569 Office Supplies Paper Xerox 232
OFF-PA-10000587 Office Supplies Paper Array Parchment Paper, Assorted Colors
OFF-PA-10003892 Office Supplies Paper Xerox 1943
OFF-PA-10000482 Office Supplies Paper Snap-A-Way Black Print Carbonless Ruled Speed Letter, Triplicate
OFF-PA-10004675 Office Supplies Paper Telephone Message Books with Fax/Mobile Section, 5 1/2" x 3 3/16"
OFF-PA-10000304 Office Supplies Paper Xerox 1995
OFF-PA-10003177 Office Supplies Paper Xerox 1999
OFF-PA-10003256 Office Supplies Paper Avery Personal Creations Heavyweight Cards
OFF-PA-10004040 Office Supplies Paper Universal Premium White Copier/Laser Paper (20Lb. and 87 Bright)
OFF-PA-10004965 Office Supplies Paper Xerox 1921
OFF-PA-10002377 Office Supplies Paper Xerox 1916
OFF-PA-10001804 Office Supplies Paper Xerox 195
OFF-PA-10001736 Office Supplies Paper Xerox 1880
OFF-PA-10004327 Office Supplies Paper Xerox 1911
OFF-PA-10001950 Office Supplies Paper Southworth 25% Cotton Antique Laid Paper & Envelopes
OFF-PA-10002254 Office Supplies Paper Xerox 1883
OFF-PA-10002751 Office Supplies Paper Xerox 1920
OFF-PA-10000659 Office Supplies Paper Adams Phone Message Book, Professional, 400 Message Capacity, 5 3/6¿½ x 11ï¿½
OFF-PA-10001144 Office Supplies Paper Xerox 1913
OFF-PA-10000061 Office Supplies Paper Xerox 205
OFF-PA-10002749 Office Supplies Paper Wirebound Message Books, 5-1/2 x 4 Forms, 2 or 4 Forms per Page

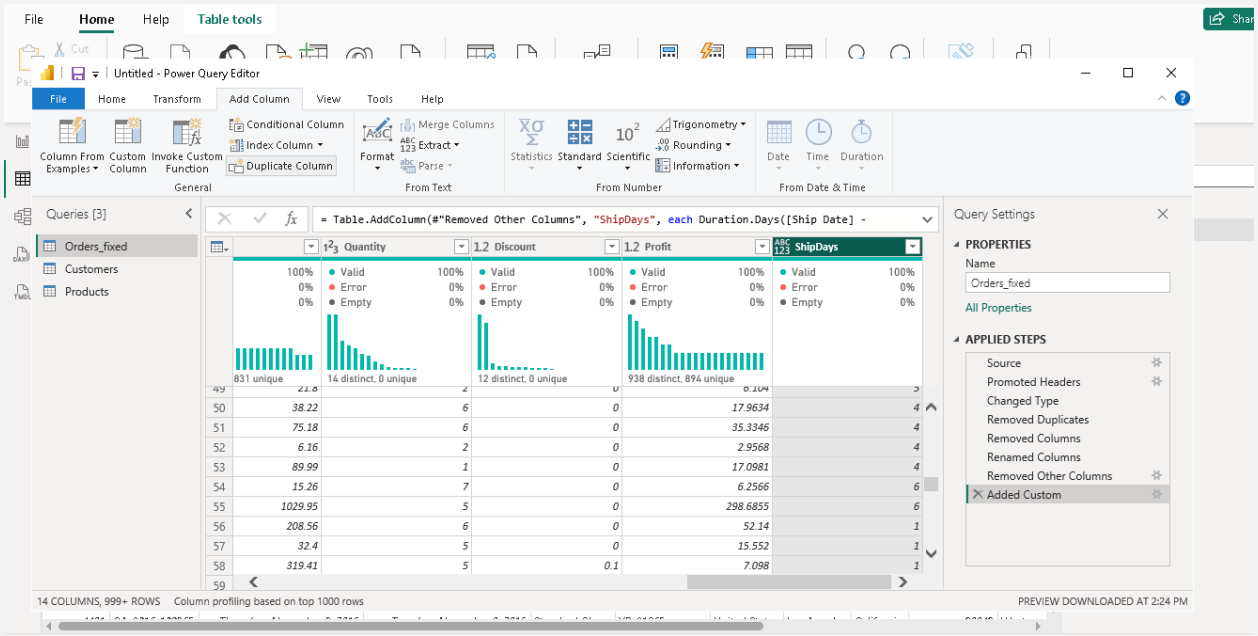
Task 2: Data Cleaning in Power Query

All cleaning was done in Power Query Editor (Home > Transform data). Every action is recorded in Applied Steps.

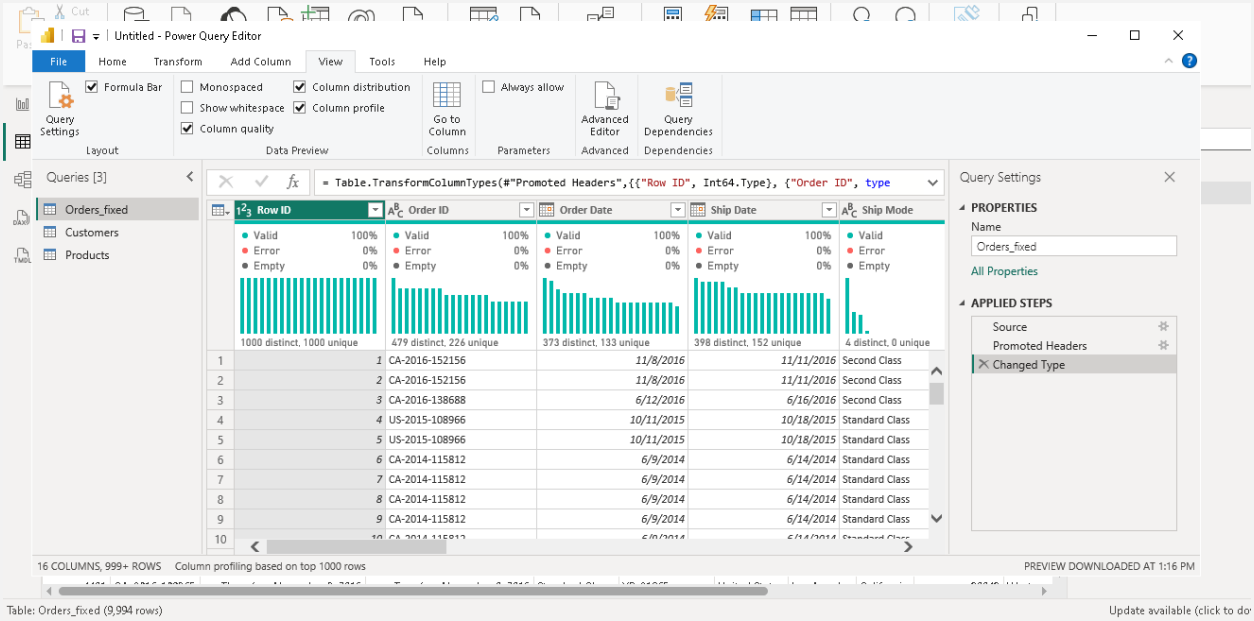
1. Promoted first row to headers (Use First Row as Headers) for Customers and Products.
2. Fixed data types: dates as Date, Sales/Profit/Discount as decimal, Quantity as whole number, IDs as text.
3. Date format problem: M/D/YYYY text dates were read wrongly, so the dates were loaded in ISO format (YYYY-MM-DD) and the error was removed.
4. Checked Column quality / Column profile for errors, empty values and duplicates.
5. Removed duplicate rows and removed columns that were not needed for analysis.
6. Created a new column ShipDays = Duration.Days([Ship Date] - [Order Date]).
7. Renamed the query to Orders and clicked Close & Apply.

```
ShipDays = Duration.Days([Ship Date] - [Order Date])
```

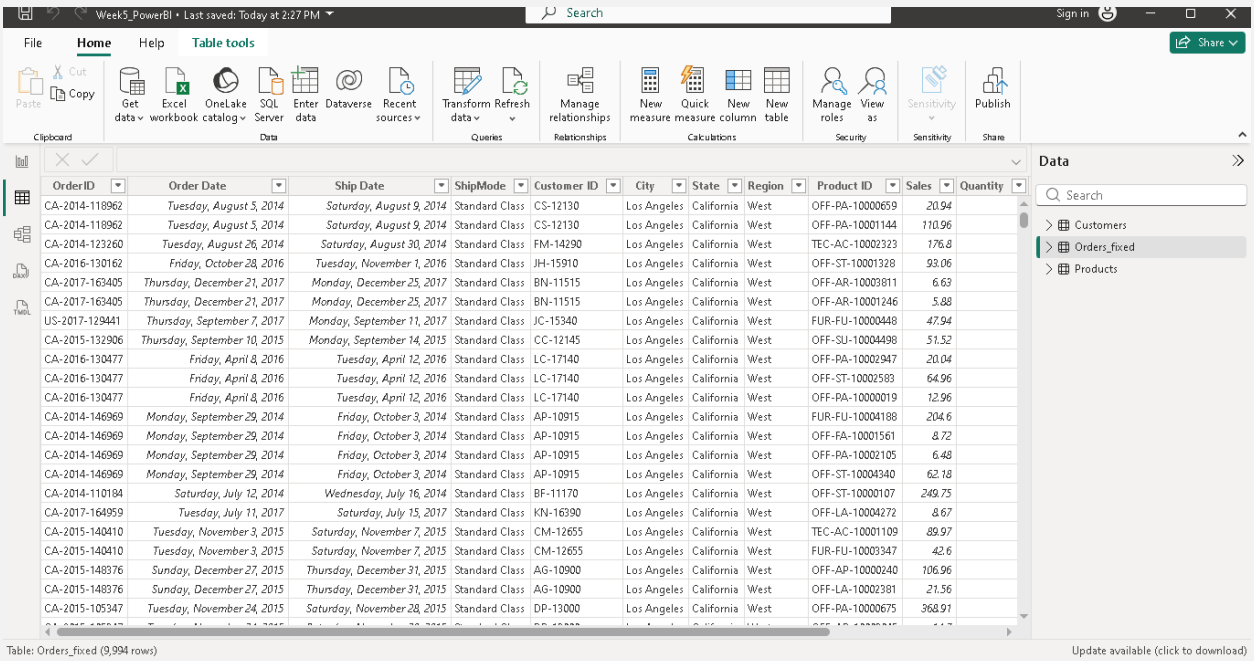
[Screenshot here: Power Query: Applied Steps for Orders]



[Screenshot here: Column quality / Column profile]



[Screenshot here: Custom column ShipDays]



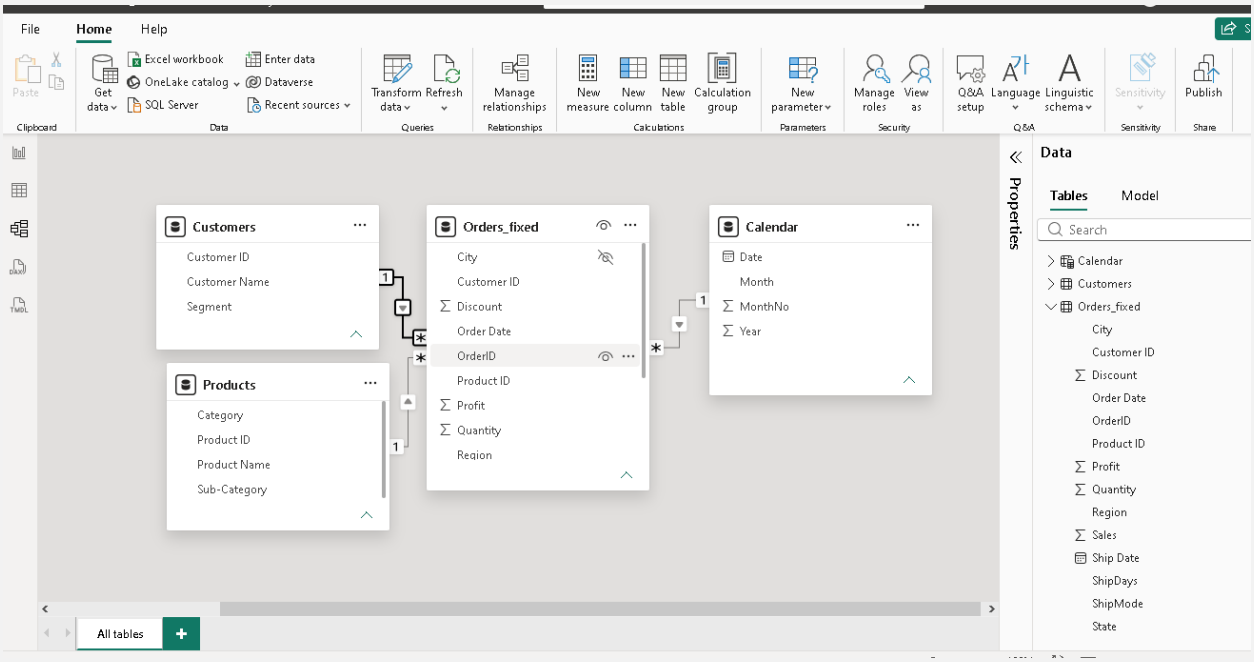
Task 3: Data Modeling

A star schema was created: Orders is the fact table, and Customers, Products and Calendar are dimension tables. All relationships are Many-to-one with single direction filtering.

From (Many)	To (One)	Type
Orders[Customer ID]	Customers[Customer ID]	Many-to-one, Single
Orders[Product ID]	Products[Product ID]	Many-to-one, Single
Orders[Order Date]	Calendar[Date]	Many-to-one, Single

Benefit: a filter on Customers, Products or Calendar (for example the Year slicer) flows to Orders, so all visuals update together.

[Screenshot here: Model view showing all four tables and relationships]



Task 4: DAX Measures & Calculated Columns

4.1 Calendar table (calculated table)

```
Calendar = ADDCOLUMNS(
    CALENDAR(DATE(2014,1,1), DATE(2017,12,31)),
    "Year", YEAR([Date]),
    "MonthNo", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"))
```

4.2 Measures

```
Total Sales = SUM(Orders[Sales])
Total Profit = SUM(Orders[Profit])
Profit % = DIVIDE([Total Profit], [Total Sales])
Average Sales = AVERAGE(Orders[Sales])
Total Records = COUNTROWS(Orders)
Total Orders = DISTINCTCOUNT(Orders[OrderID])
Max Sale = MAX(Orders[Sales])
Min Sale = MIN(Orders[Sales])
```

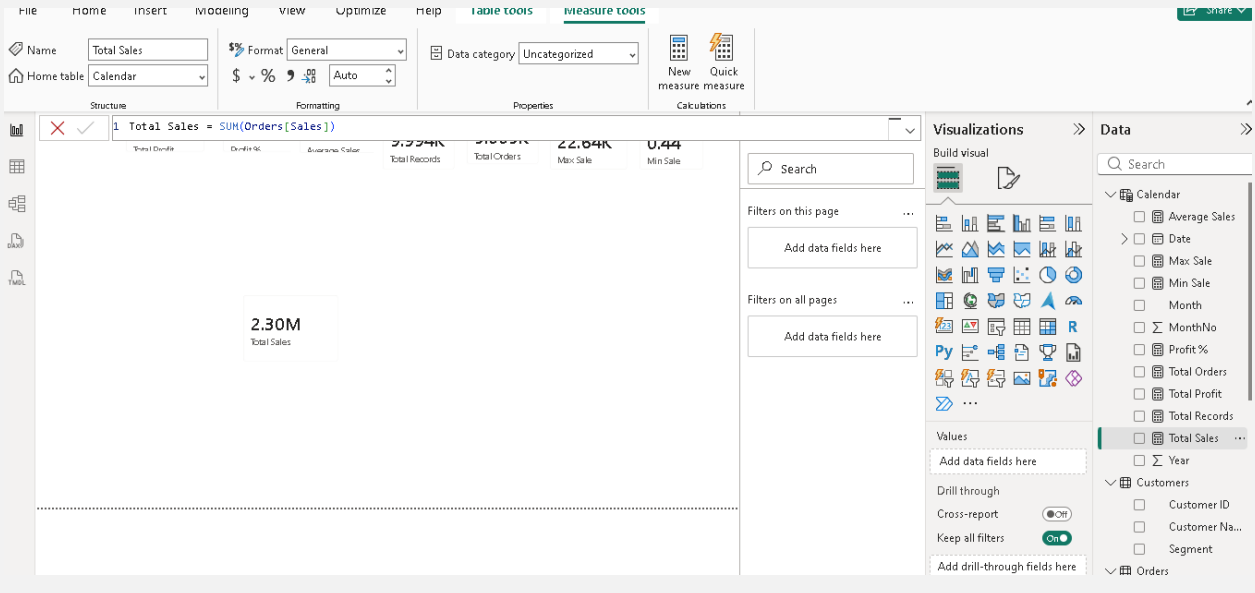
4.3 Calculated column

```
Profit Status = IF(Orders[Profit] < 0, "Loss", "Profit")
```

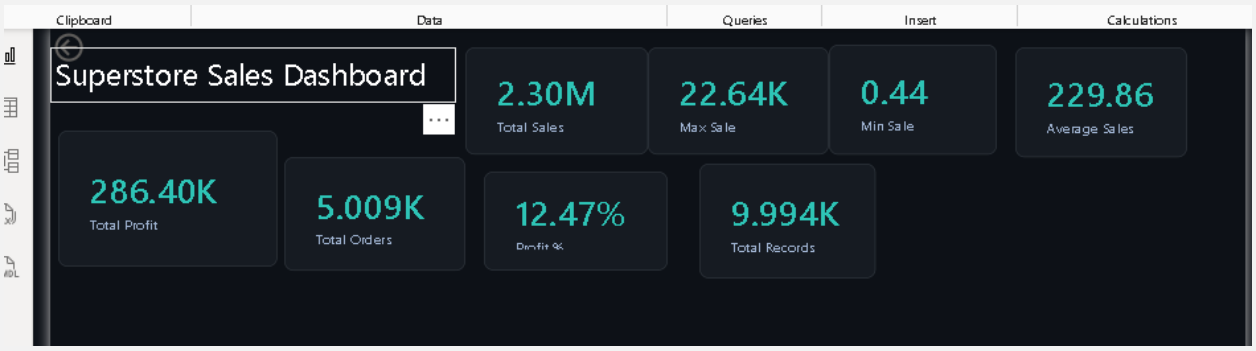
4.4 Results

Measure	Result
Total Sales	2.30M
Total Profit	286.40K
Profit %	12.47%
Average Sales	229.86
Total Records	9,994
Total Orders	5,009
Max Sale	22,638.48
Min Sale	0.44

[Screenshot here: DAX measures in the formula bar]



[Screenshot here: Measures shown in a Card / Table visual]

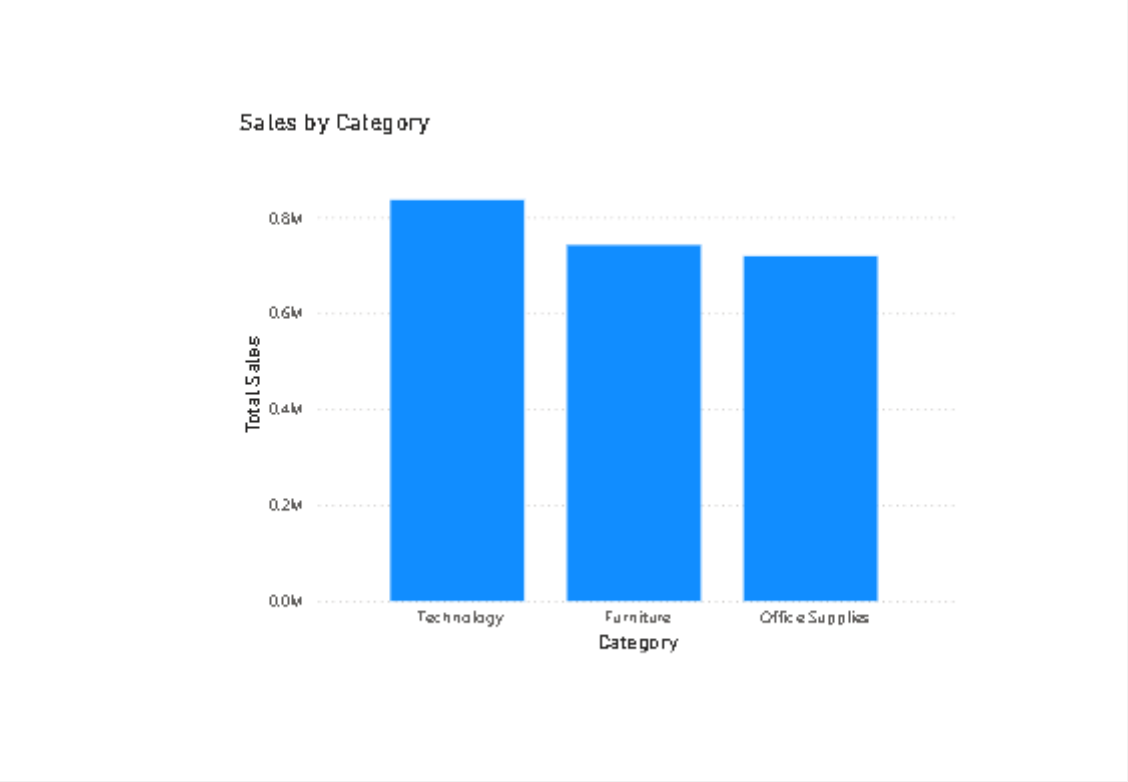


Task 5: Visualizations

5.1 Column chart: Sales by Category

Axis: Category, Values: Total Sales. Technology has the highest sales.

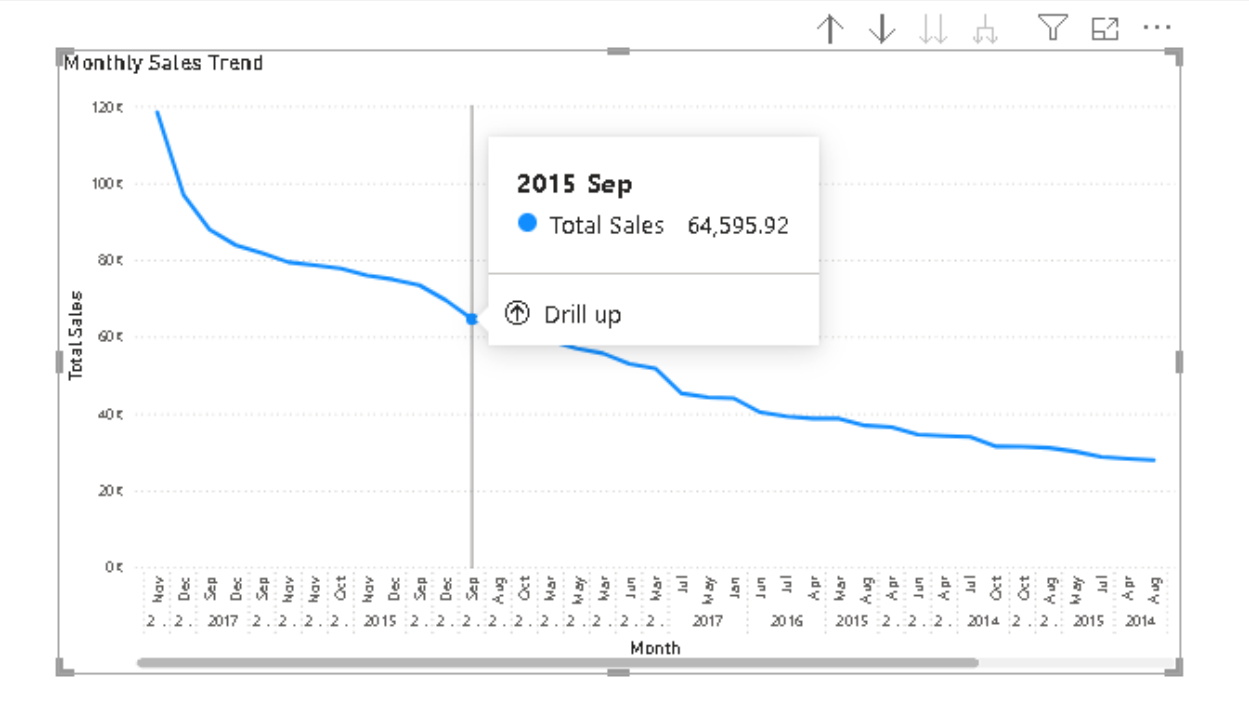
[Screenshot here: Column chart: Sales by Category]



5.2 Line chart: Monthly Sales Trend

Axis: Month, Values: Total Sales. Shows how sales change across months.

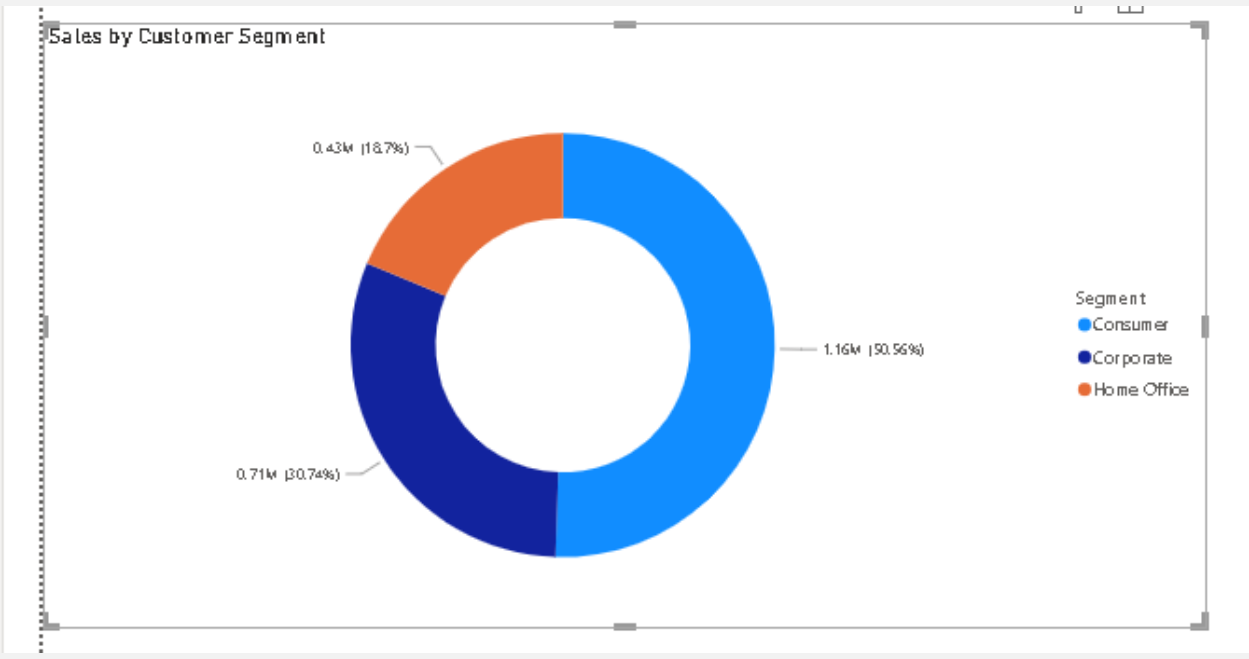
[Screenshot here: Line chart: Monthly Sales Trend]



5.3 Donut chart: Sales by Customer Segment

Legend: Segment, Values: Total Sales. Consumer has 50.56% share.

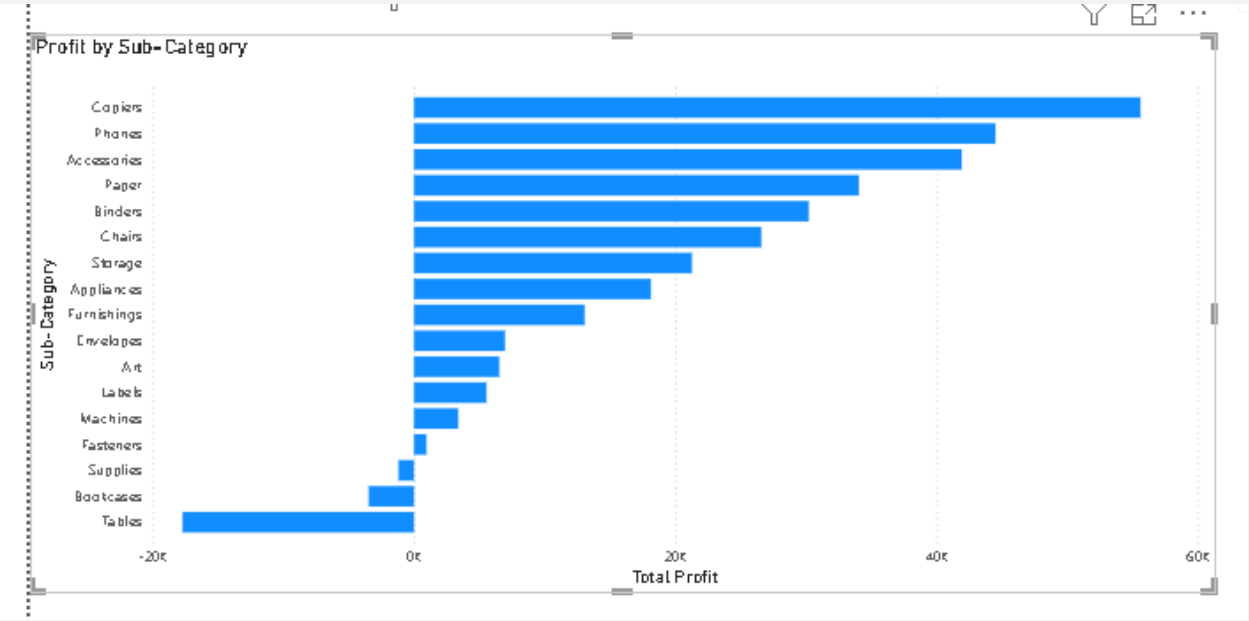
[Screenshot here: Donut chart: Sales by Customer Segment]



5.4 Bar chart: Profit by Sub-Category

Axis: Sub-Category, Values: Total Profit. Copiers give the highest profit.

[Screenshot here: Bar chart: Profit by Sub-Category]



5.5 Table: Top Products

Product Name with Total Sales and Total Profit, sorted by Total Sales (high to low).

[Screenshot here: Table: Top Products]

Top Product by Sales

Product Name	Total Sales	Total Profit
Canon imageCLASS 2200 Advanced Copier	61,599.82	25,199.93
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27,453.38	7,753.04
Cisco TelePresence System EX90 Videoconferencing Unit	22,638.48	-1,811.08
HON 5400 Series Task Chairs for Big and Tall	21,870.58	0.00
GBC DocuBind TL300 Electric Binding System	19,823.48	2,233.51
GBC Ibimaster 500 Manual ProClick Binding System	19,024.50	760.98
Hewlett Packard LaserJet 3310 Copier	18,839.69	6,983.88
HP Designjet T520 Inkjet Large Format Printer - 24" Color	18,374.90	4,094.98
GBC DocuBind P400 Electric Binding System	17,965.07	-1,878.17
High Speed Automatic Electric Letter Opener	17,030.31	-262.00
Lexmark MX611dhe Monochrome Laser Printer	16,829.90	-4,589.97
Martin Yale Chadless Opener Electric Letter Opener	16,656.20	-1,299.18
Ibico EPK-21 Electric Binding System	15,875.92	3,345.28
Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	15,610.97	-669.54
3D Systems Cube Printer, 2nd Gereration, Magenta	14,299.89	3,717.97
Samsung Galaxy Mega 6.3	13,943.67	1,822.76
Logitech G19 Programmable Gaming Keyboard	13,756.54	4,425.34
Apple iPhone 5	12,996.60	1,611.58
Bretford Rectangular Conference Table Tops	12,995.29	-327.23
Global Troy Executive Leather Low-Back Tilter	12,975.38	951.86
DMI Eclipse Executive Suite Bookcases	12,921.64	19.69
Canon PC1060 Personal Laser Copier	11,619.83	4,570.93
SAFCO Arco Folding Chair	11,572.78	1,179.37
Honeywell Enviracaire Portable HEPA Air Cleaner for 17' x 22' Room	11,304.44	3,247.02
Total	2,297,200.86	286,397.02

Task 6: Interactive Dashboard

The dashboard "Superstore Sales Dashboard" uses a dark theme with a consistent colour palette (teal, blue, orange) so that it is easy to read.

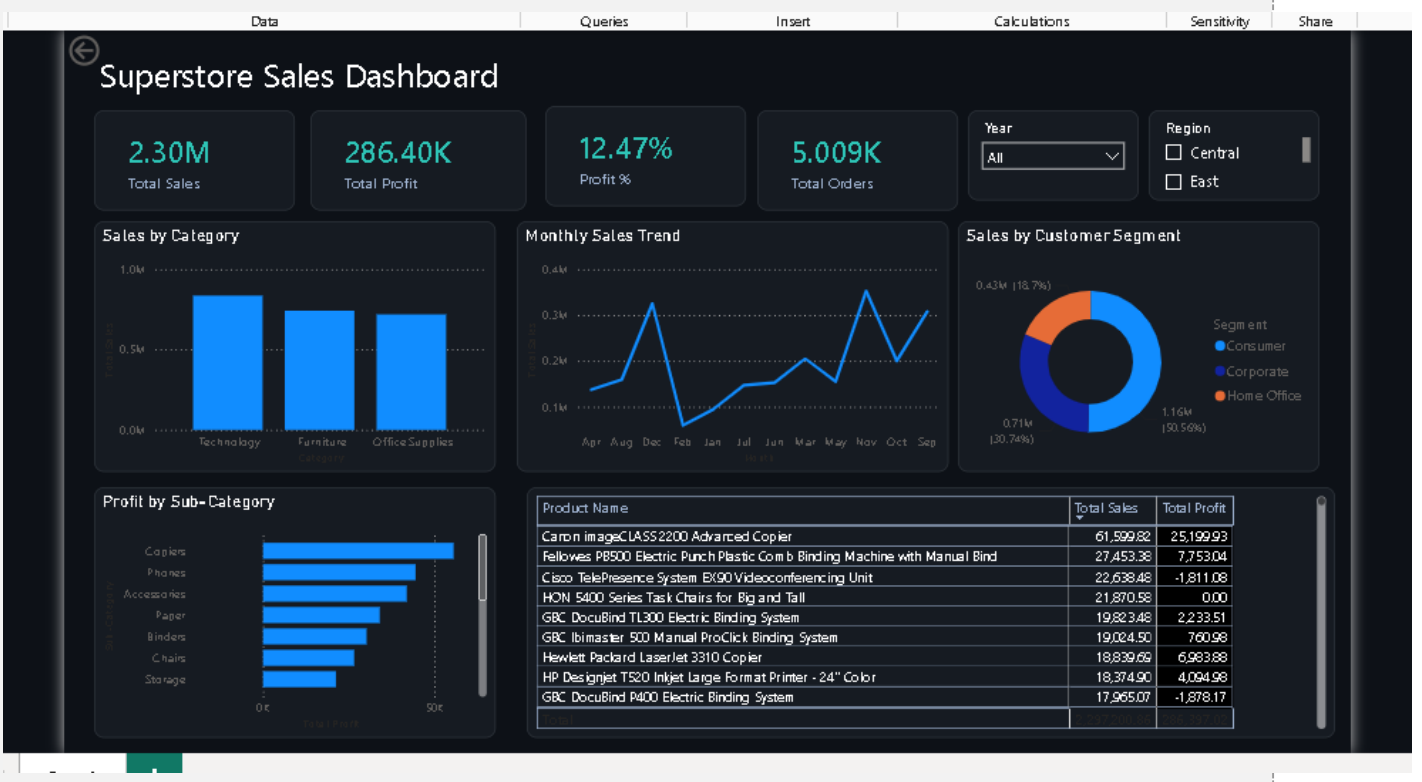
Contents

- 4 KPI cards: Total Sales (2.30M), Total Profit (286.40K), Profit % (12.47%), Total Orders (5,009).
- 5 visuals: Sales by Category, Monthly Sales Trend, Sales by Customer Segment, Profit by Sub-Category, Top Products table.
- 2 slicers: Year and Region.

Interactivity

When a slicer value is selected, all cards and visuals update together. Clicking a bar or a donut slice also cross-filters the other visuals.

[Screenshot here: Full dashboard (no filter)]



[Screenshot here: Dashboard with Year = 2017 and Region = West]



Task 7: Insights & Recommendations

Key Insights

1. The Consumer segment is the largest, with about 50.56% of sales (1.16M). Corporate has 30.74% and Home Office 18.7%.
2. Technology is the top category by sales. Furniture and Office Supplies are close to each other.
3. Copiers give the highest profit among sub-categories. The Canon imageCLASS 2200 Advanced Copier is the top product (Sales 61,599.82, Profit 25,199.93).
4. Some high-selling products make a loss, for example Cisco TelePresence System EX90 (-1,811.08) and GBC DocuBind P400 (-1,878.17).
5. The overall profit margin is 12.47%, i.e. about 12 rupees profit on every 100 rupees of sales.
6. Monthly sales go up in the last quarter of the year (September to December), which points to seasonal demand.

Recommendations

1. Reduce discounts or revise the price of loss-making products such as Cisco TelePresence and GBC DocuBind P400.
2. Give more focus (stock and promotion) to Technology and Copiers because they bring the most profit.
3. Prepare stock and offers before September to December to use the seasonal demand.
4. Launch special offers for Corporate and Home Office customers to grow their share.

Conclusion

Power BI helped to convert raw sales data into a clean model and an interactive dashboard that shows sales, profit, customers and products in one place and supports better business decisions.